

A prior $p(\boldsymbol{\theta}) \in \mathcal{F}$ is a conjugate prior for a likelihood function $p(\mathbf{y}|\boldsymbol{\theta})$ if the posterior $p(\boldsymbol{\theta}|\mathbf{y})$ is in the same family $\mathcal{F}$ as the prior. In other words, if the prior and likelihood are conjugate, then the posterior is also in the same family. To note, if the family $\mathcal{F}$ corresponds to the [Exponential Family] of distributions, then the computation of the posterior is particularly straightforward, achieving a closed-form solution.

0.0.1 See also

0.0.2 References